

E-Ink – The Flagbearer

The E-Ink Corporation, as mentioned earlier, was founded by the same people who invented the microencapsulated electrophoretic display. The company, post an acquisition and years of creating and improving various types of e-paper displays, consists of a major share of the market right now. There are a number of e-Ink solutions available, each with distinctive features and target usage. They are as follows:

- ◆ **Pearl:** With Pearl there was a significant improvement upon the existing technology with a much better contrast resembling a paper book, along with a 50-250 ms refresh range. With the Pearl solution, E-Ink offers support from capacitive and digitizer touchscreens. The digitizer layer is below the TFT and requires a stylus, whereas the capacitive technology is located at the top of the display to detect finger swipes. Some of the most significant devices to use this type of display are the Amazon Kindle DX, Kindle 3, Kindle 4, Kindle Touch, Nook Simple Touch and Kobo eReader Touch.
- ◆ **Carta:** Announced in the January 2013 CES, Carta offered increased readability by an almost 50 per cent improvement along with a resolution of 212 pixels per inch (ppi). It also incorporated E-Ink's new Regal technology which allowed partial page refreshes allowing much smoother page turns. Carta has been used in the Kindle Paperwhite 1st and 2nd Gen devices. A later edition called Carta HD, supporting 300 ppi, was used in Kindle Voyage, Kindle Paperwhite 3rd Gen, Kobo eReader Aura H2O and Kobo Glo HD.
- ◆ **Spectra:** At the same generation level as Pearl, Spectra allowed 3 colours – black, white as well as a third colour based on a third pigment inside the microcapsule. Potential usage was in shopping tags and dynamic price changing on products while keeping them attractive enough to grab the attention of shoppers with a dash of colour.



The Amazon Kindle DX with the E-Ink Pearl Display